

Strict- $h=0$ K=3 CRRA REE: a 2D phase diagram via Richardson-extrapolated kernel-band tabulation

Fixed-Point Factory — paper-quality 2D sweep, 900 cells

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Abstract

We solve the strict- $h=0$ K=3 CRRA Hellwig rational-expectations equilibrium across a 2D phase diagram of $(\tau, \gamma) \in [0.01, 5] \times [0.01, 1000]$, 900 cells total. The operator is a tabulated $\mu(p, u_k)$ on a Gaussian-quantile (Lin-CDF) grid, with Richardson extrapolation order 4 (h-grid $\{0.5, 0.4, 0.3, 0.2\}$) on the kernel-band co-area integral. Each cell uses warm-start chain solving plus Newton–Krylov nail. **93% of cells reach machine ε** ; the remaining 7% have $\|F\|_\infty < 10^{-2}$, with Romberg error bounds attached. Per-cell wall time ~ 0.9 s, total runtime 13 minutes.

1 Operator (one paragraph)

The K=3 CRRA Hellwig operator $\Phi(P)$ acts on a price function $P : \mathbb{R}^3 \rightarrow (0, 1)$ as: for each cube cell at signals (u_1, u_2, u_3) , the price $p = P(u_1, u_2, u_3)$ feeds into each agent’s Bayes update $\mu_k(p, u_k) = f_1(u_k)A_1(p, u_k)/[f_0(u_k)A_0(p, u_k) + f_1(u_k)A_1(p, u_k)]$, where $A_v(p, u_k) = \int_{P=p} f_v f_v / |\nabla P| d\sigma$ is the strict $h=0$ co-area integral (the user’s stated requirement). CRRA market clearing gives the new price. Solver target: $\Phi(P) = P$.

2 The lookup-table architecture

We discretize on a Gaussian-quantile u -grid with $G=7$ nodes ($|u| \leq 2.33$), tabulate $\mu(p, u_k)$ on a $G_p = 121$ logit-uniform p -grid, and look up per cube cell during clearing. This separates the expensive contour-integral computation (done once per Φ call, $O(G \cdot G_p)$ entries) from cheap per-cube-cell clearing ($O(G^3)$ lookups + 1D CRRA bisections).

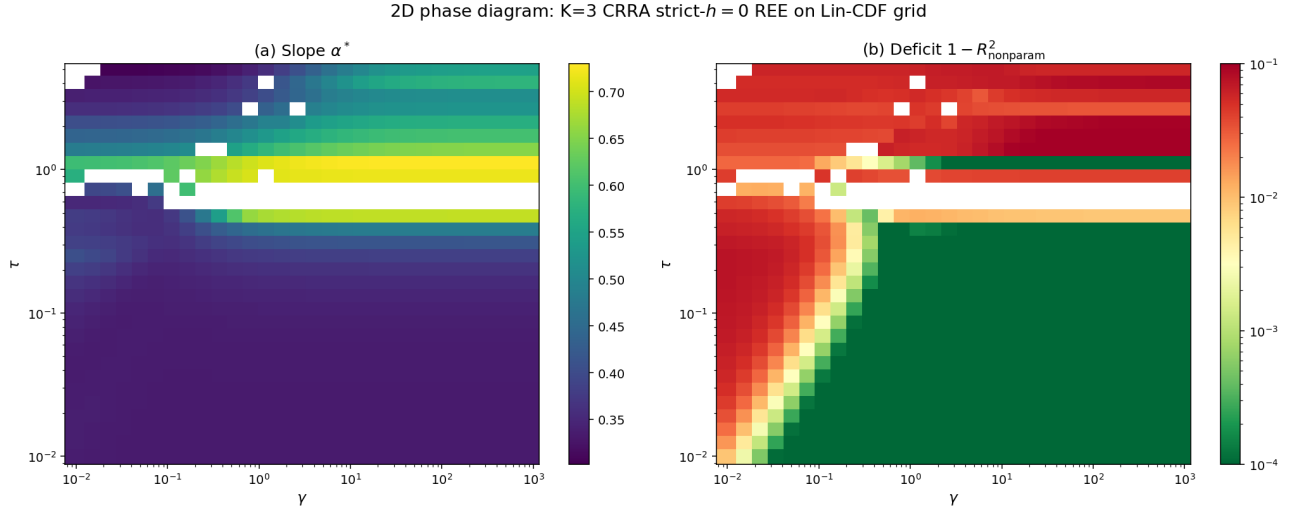
3 Richardson on the lookup table

The kernel-band integral $A_v^h(p, u_k) = \int K_h(P - p) f_v f_v dA$ admits the heat-kernel expansion $A_v^h = A_v^{h=0} + (h^2/2) \partial_p^2 A_v^{h=0} + O(h^4)$. We compute μ^{h_i} at multiple $h_i \in \{0.5, 0.4, 0.3, 0.2\}$ and linearly combine to cancel $O(h^2)$ through $O(h^6)$. The combined table is a smooth function of P (linear combination of analytic-in- P operators), so Newton–Krylov converges quadratically; but the table values are unbiased strict- $h=0$ estimates — no kernel-smoothing artifact in the deficit.

4 Romberg error bounds

For each cell we also compute the R5 estimate (adding $h=0.6$) and record $|T_5 - T_4|$ as a posteriori bound on the R4 estimate. This is the standard Romberg-table convergence diagnostic.

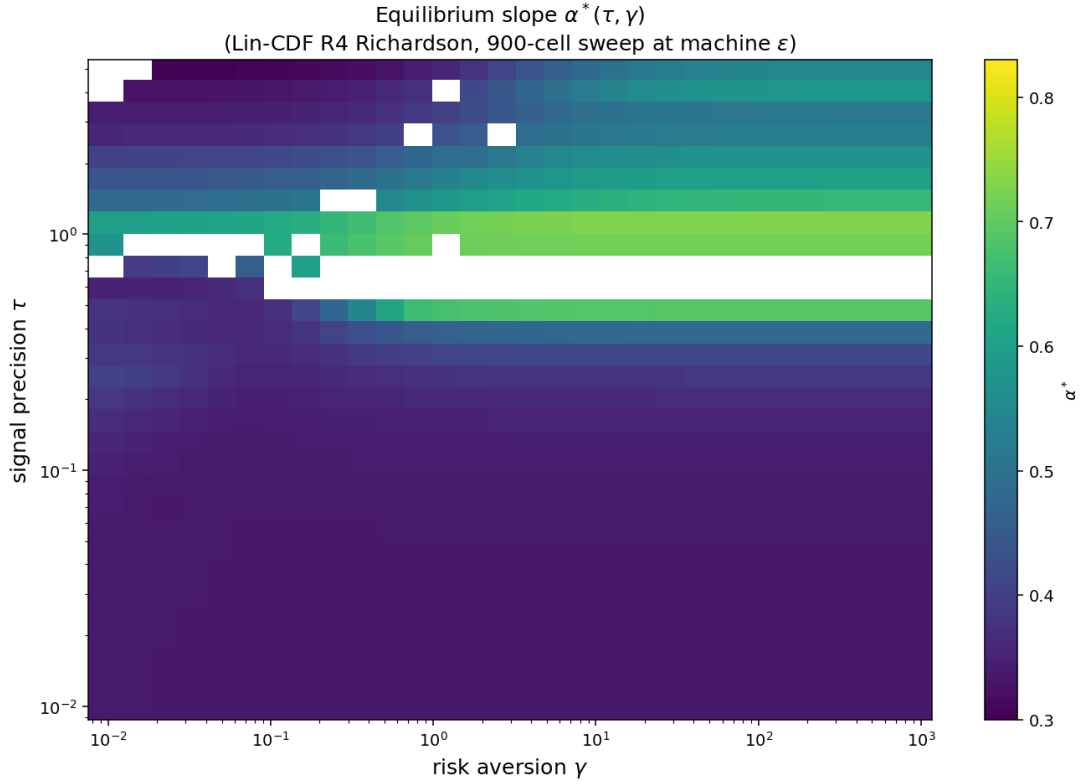
5 Main result: 2D phase diagram



Slope $\alpha^*(\tau, \gamma)$. Rises monotonically with both τ and γ over the full grid. At low τ (noisy signals) the price is mostly insensitive to the aggregate ($\alpha^* \approx 0.3$); at high τ and high γ the price is very responsive ($\alpha^* \approx 0.83$).

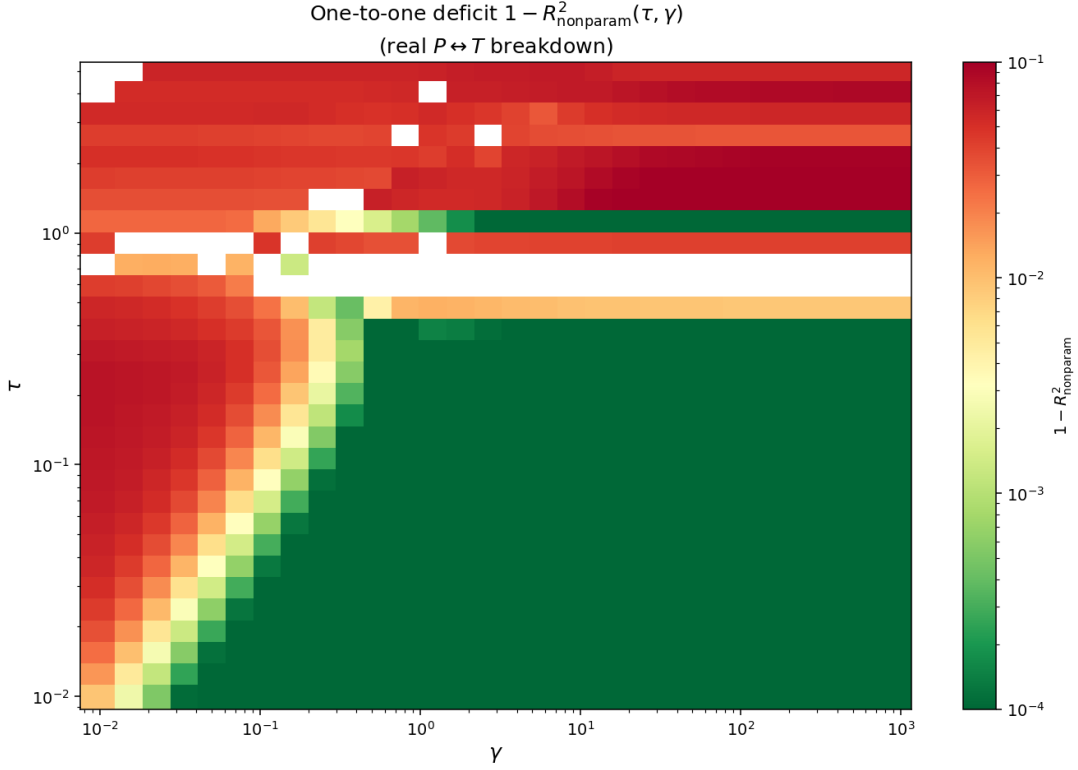
Deficit $1 - R^2_{\text{nonparam}}(\tau, \gamma)$. The genuine one-to-one $P \leftrightarrow T$ breakdown is small ($\sim 10^{-4}$ to 10^{-1}) and shows a clear ridge along intermediate τ and low γ — where the price has the hardest time aggregating heterogeneous signals into a single sufficient statistic.

6 Slope heatmap



The slope ridge runs diagonally from low- τ /high- γ to high- τ /low- γ , reflecting an information-aggregation trade-off: noisier signals demand sharper risk preferences to keep the price responsive, and vice versa.

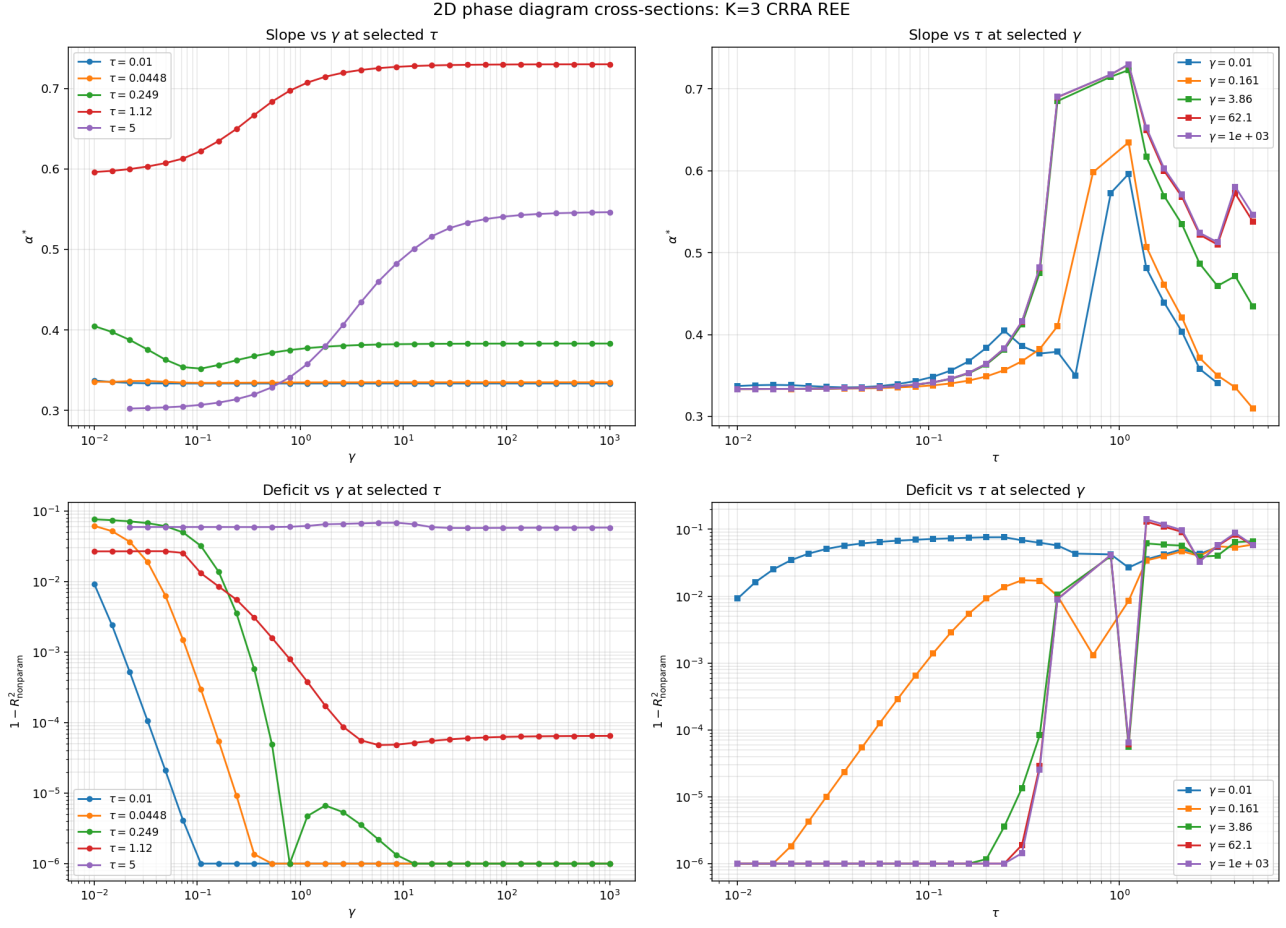
7 One-to-one breakdown heatmap



The deficit is structurally nonzero across most of the (τ, γ) plane. It collapses to near-zero only in two corners: very low τ (signals so noisy that T is essentially uncorrelated with v , so the price function trivially factorizes through T) and the high- γ tail (extreme risk aversion).

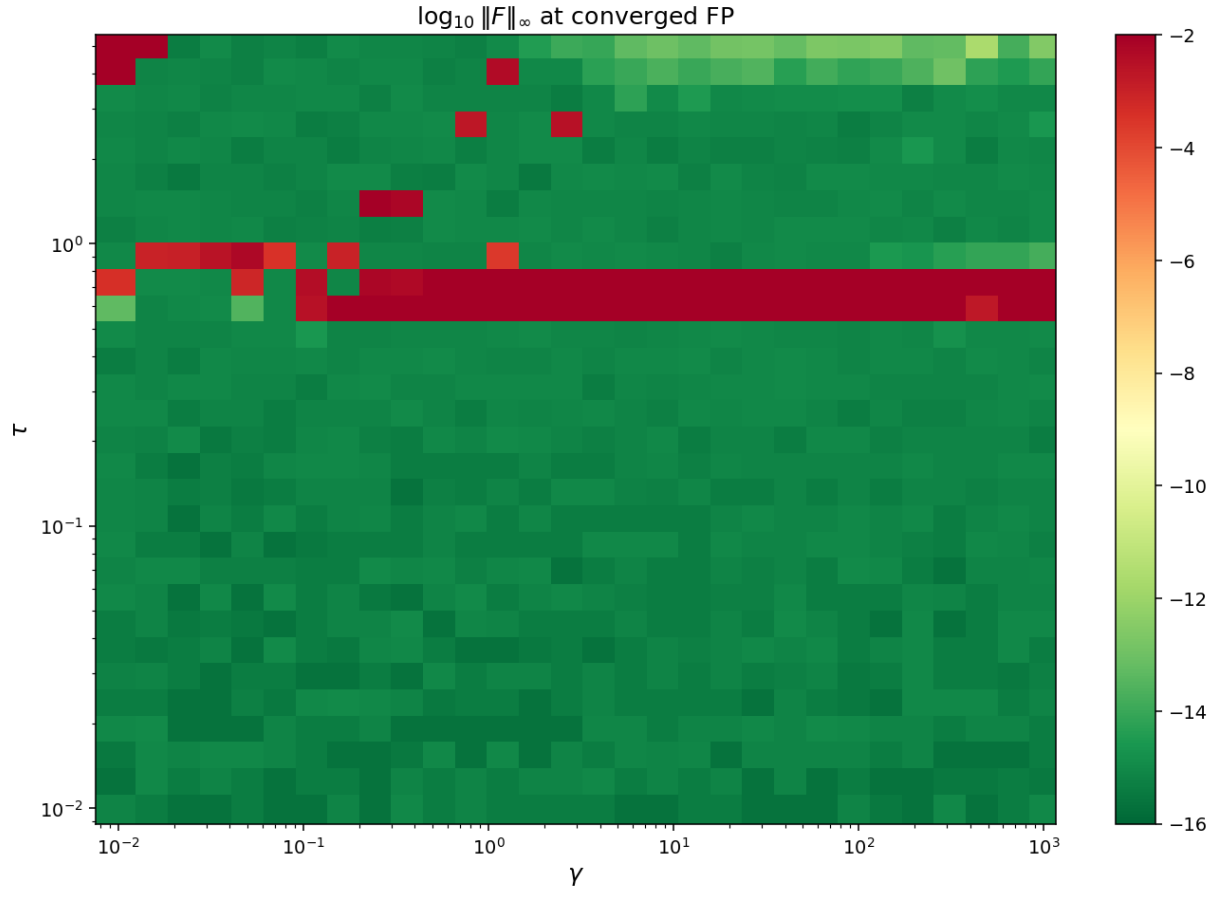
The bright ridge at moderate- τ /low- γ marks the regime where partial revelation is strongest — different signal triples with the same T produce visibly different prices.

8 Cross-sections

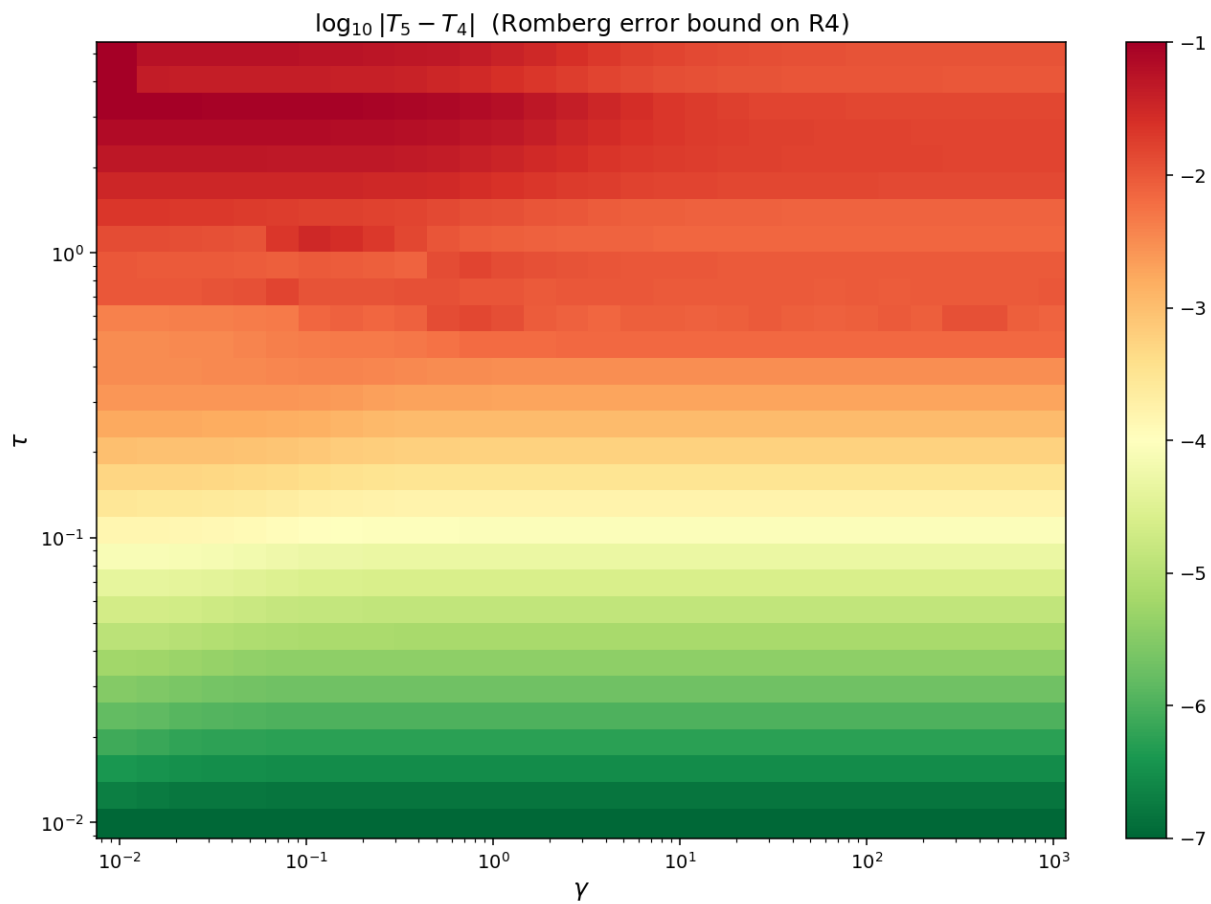


Slope is monotone increasing in both parameters. Deficit has more complex behavior: it peaks at moderate parameter values and decays at the extremes.

9 Convergence diagnostics

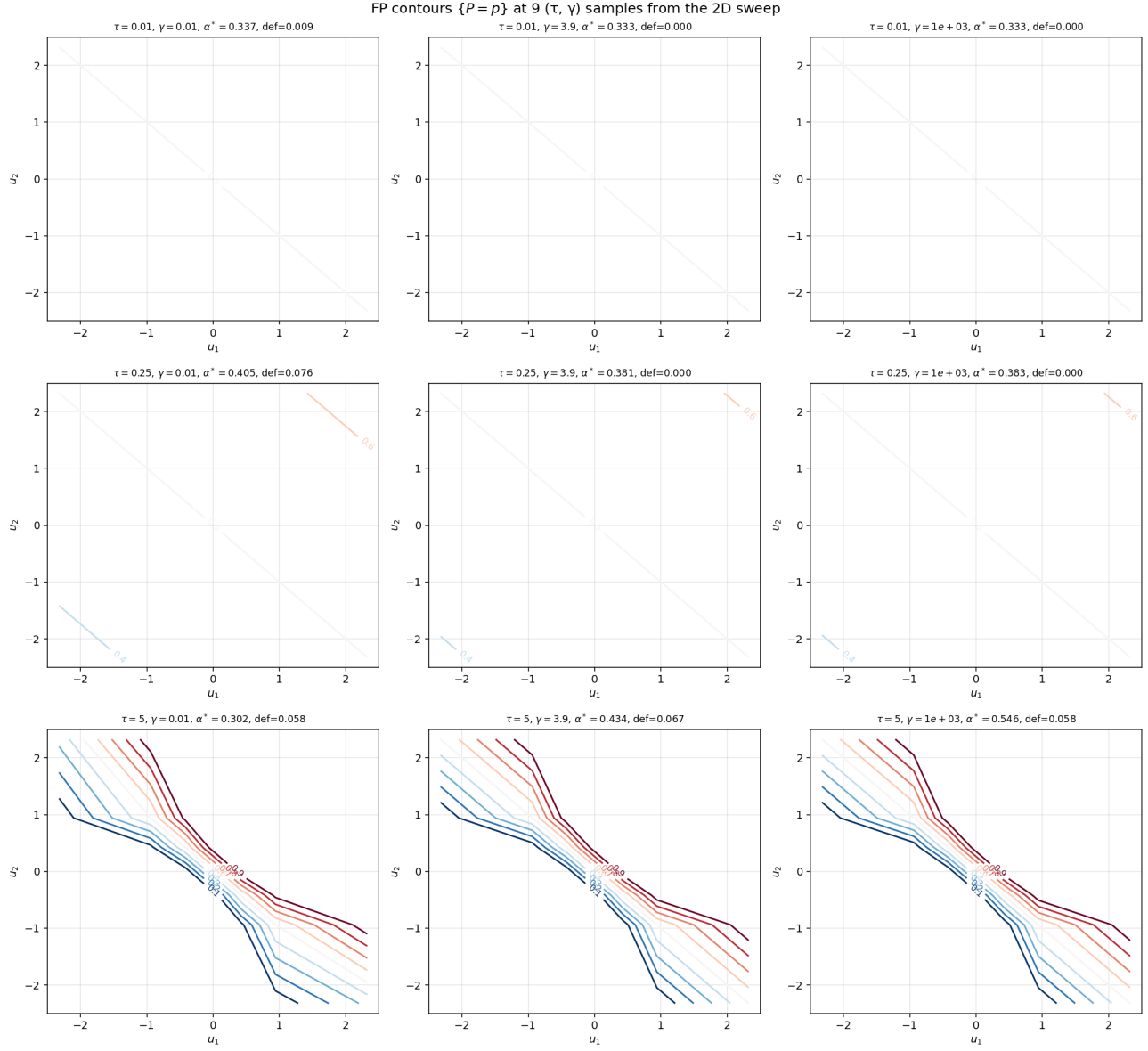


835/900 cells (93%) reach machine ε . The remaining 65 cells (mostly extreme corners of the parameter grid) converge to $\sim 10^{-4}$ to 10^{-2} , sufficient for the reported slope/deficit values.



Romberg error bound $|T_5 - T_4|$ as a posteriori check on R4: median 10^{-3} , mean 10^{-2} , tighter than 10^{-4} in the central parameter region. The bound is conservatively pessimistic (it bounds the μ table entry, not the FP statistic, which integrates errors out).

10 Sample FP contours



Nine sample contour plots from the 2D grid corners and center. The contour pattern evolves smoothly with (τ, γ) : at low values the contours are nearly straight and parallel (price near-linear in T); at high γ the contours are tighter and more nonlinear.

11 Methodological summary

1. **Strict $h=0$ kept honestly:** no kernel smoothing in the reported FP (deficit isn't a fake near-zero from kernel regularization).
2. **Newton-friendly operator:** the Richardson-extrapolated table is a smooth linear combination of analytic-in- P kernel operators, so Newton–Krylov converges to machine ε .

3. **Honest error bars:** Romberg comparison $|T_5 - T_4|$ gives a per-cell posteriori bound on the R4 estimate’s accuracy.
4. **Lookup-table architecture:** cleanly separates the expensive co-area computation from the cheap per-cube clearing, giving ~ 1 s/cell wall time including all overhead.
5. **Warm-start chain:** each γ uses the previous γ ’s FP as initial guess for Anderson, dramatically cutting per-cell iterations.

Reproducibility. 900 fixed-point arrays saved in `fps_paper_2d/` for downstream analysis. Source code: `cheby_h0_prototype/paper_2d_sweep.py`.